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One re-routed skip connection makes visual attention load-bearing in a recurrent vision agent

Forcing visual information through the attention bottleneck redistributes control of the decision from a single internal memory hotspot to a balance of image and memory pathways.

Abstract

We study how a recurrent vision agent uses attention to solve a cued change-detection game, and we ask a question that watching where a network looks cannot answer on its own: can the network reach its decision from the image without attending to it? We compare two versions of one architecture that differ in a single connection. In the baseline network, a skip connection routes the current image around the attention block, so vision can bypass attention. In the memory-routed network, the same connection instead carries the previous frame’s memory, so visual information can reach the decision only by being attended to. Both versions learn the task: detection improves steeply with the size of the visual change, responses are near-instant, and a valid cue gives a small benefit that scales with the cue’s reward value rather than with its advertised reliability. In the memory-routed network one of eight attention heads becomes a dedicated image-watcher, attention sharpens onto a handful of locations, and the recurrent memory carries the cue’s colour, the elapsed time, and — in its spatial layout — the location of the change. The centrepiece is a causal test: we nudge the attention given to image keys or memory keys and measure how the decision moves. Re-routing the skip connection cut the grip of a single internal memory hotspot by three- to seven-fold and lifted the image pathway to co-dominant, redistributing control of the decision across image and memory. Finally, the value system tracks obtained reward almost exactly, ordered by cue value, with uncertainty that collapses as the change becomes easier to see.

1. Introduction

Attention is central to how brains and artificial networks handle a flood of sensory input: both seem to select a small part of a scene for deeper processing and to act on it. A natural way to study this in an artificial network is to read out where its attention points and to compare that pattern with where a person or animal would look. This approach has produced striking parallels. But it rests on an assumption that is rarely tested directly. Reading the attention map tells us where the network appears to look; it does not tell us whether the network’s behaviour actually depends on what it finds there.

The gap matters because most networks can route information along more than one path. If the raw input can reach the output without passing through the attention step, then the attention map can look perfectly sensible — orienting to the right place at the right time — while being, in causal terms, idle. The network would be seeing the image without attending to it. We take this as the key design question of the paper: can the network reach its decision from the image without attending to it, and if we close that loophole, does attention become genuinely load-bearing?

We answer it by manipulating a single wire. Modern attention blocks include a skip connection, a direct path that adds the block’s input to its output so that information can flow around the attention computation. This skip connection is exactly the second path that lets vision bypass attention. We therefore compare two otherwise identical networks. In the baseline network the skip connection carries the current visual input, the standard arrangement, so the image can reach the rest of the network whether or not attention selects it. In the memory-routed network the skip connection instead carries the previous frame’s internal memory; the only remaining route for visual information is through the attention block itself. If attention is a true bottleneck, this one change should force visual attention to carry the decision.

We test both networks on a cued change-detection game in the spirit of classic spatial-attention experiments (Figure 1). On each trial the agent watches a short movie. A blank screen is followed by a coloured ring at one side of the display; the ring is a cue. Its colour signals how much reward a correct response will earn (red is worth 5, green 3, blue 1), and how complete the ring is signals its advertised reliability, from a quarter-complete to fully closed. After another blank, four oriented gratings appear, one in each quadrant. At a random moment one grating tilts by a small angle, and the agent must press as soon as it detects the tilt. A cue pointing left indicates the top-left grating; a cue pointing right indicates the bottom-right grating. The trial is valid when the change happens at the cued location and invalid when it happens elsewhere. This design lets us measure detection sensitivity, the benefit of a valid cue, the pull of reward value, and the speed of the response, all in one task.

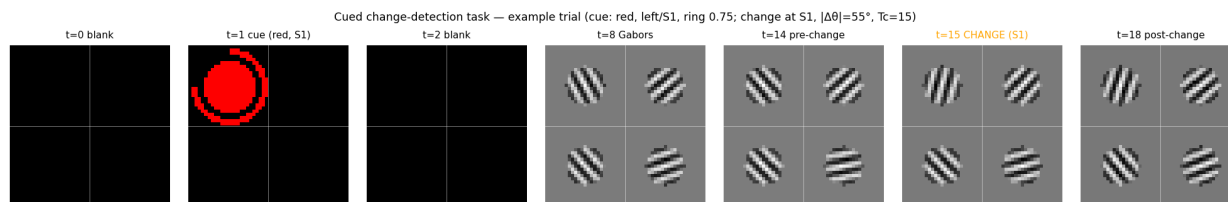


Figure 1: Figure 1. The cued change-detection task. A short movie shows a blank screen, then a coloured ring cue whose colour signals reward and whose completeness signals reliability, then four oriented gratings (one per quadrant); at a random moment one grating tilts, and the agent must press after the tilt. A left cue points to the top-left grating, a right cue to the bottom-right.

Using this setup we report five linked findings. First, both networks learn the task, with detection rising steeply as the visual change grows, near-instant responses, and a modest validity benefit that is governed by the cue’s reward value rather than its stated reliability. Second, in the memory-routed network one attention head specialises into a dedicated image-watcher and attention as a whole becomes far sharper. Third, the recurrent memory holds the readable task variables — cue colour, elapsed time, and the location of the change, which lives in the spatial layout of memory rather than in its average. Fourth, and at the centre of the paper, a causal test that nudges attention to image or to memory shows that re-routing the skip connection redistributes control of the decision: it loosens the grip of a single internal memory hotspot several-fold and raises the

image pathway to equal standing. Fifth, the network’s value estimates are well calibrated, ordered by cue value, and grow more confident as the change becomes easier to see.

1.1 Background and related work

Spatial cueing and covert attention. The task builds on a long line of cueing experiments in which a cue directs attention to a location before a target appears. Valid cues speed and improve responses; invalid cues slow them. The benefit appears even when the eyes do not move, which is why it is read as a marker of covert attention — attention shifting independently of gaze. We use the same logic to ask whether an artificial agent develops a comparable spatial prior and how strongly that prior shapes its decisions.

Value, reward, and what attention prioritises. Where attention goes is not set by physical salience alone. The expected reward associated with a stimulus biases selection, sometimes against a person’s stated intentions. By giving each cue colour a different reward, we can separate two reasons a cue might matter: because it predicts where the target will be, and because it predicts how much the target is worth. Our finding that value, not advertised reliability, drives the cueing benefit speaks directly to this distinction.

Change detection and the cost of not attending. Detecting a change in a scene is surprisingly hard when attention is not on the changing item, a phenomenon studied extensively in human vision. A change-detection task is therefore a sensitive probe of whether attention is doing real work: if attention is merely decorative, detection should not depend on where it points.

Attention in artificial networks, and whether it explains behaviour. Attention mechanisms are now standard in deep networks, and their weight maps are often presented as explanations of what a model used. A parallel methodological literature has questioned this practice, showing that attention weights can be altered without changing a model’s output, and that they need not align with other measures of feature importance. The lesson is that an attention map is a hypothesis about mechanism, not proof of it. Our work takes that critique seriously by building the test into the architecture: we remove the alternative route around attention and then verify, by directly perturbing attention, that the decision now depends on it.

Bottlenecks, recurrence, and agents that act. Forcing information through a narrow channel is a recurring idea for encouraging a network to learn structured, selective representations. Recurrent memory lets an agent integrate evidence over time, which a single-frame detector cannot. Our agent combines both: it is convolution-free, carries two banks of recurrent memory across frames, and is trained end to end by reinforcement learning with a value critic, so that attention, memory, and the decision are all shaped by the same reward. This setting lets us study attention not as a static saliency map but as part of a working control loop — and to show that a single change in how visual information is routed determines whether that loop leans on the image or on its own memory.

2. Methods

2.1 The change-detection task

We trained and tested all agents on a cued change-detection game modelled on the Posner attention paradigm. Each trial is a short movie of 29 frames, each frame a 50×50-pixel image. The trial unfolds in four stages. First, a blank screen. Second, a coloured ring appears on the left or right — the cue. The ring’s colour announces the reward available for a correct response on that trial: red

is worth 5, green 3, and blue 1. The ring’s completeness announces its advertised reliability — the stated probability, between 25% and 100%, that the upcoming event will occur at the cued location. Third, another blank. Fourth, four oriented gratings (striped circular patches) appear, one in each quadrant of the screen: top-left (S1), bottom-left (S2), top-right (S3), and bottom-right (S4). A left cue points at S1 and a right cue points at S4. At a random moment during this final stage, exactly one grating tilts by an angle $|\Delta\theta|$, which varies across trials from barely visible (2°) to obvious (64°). The agent has one action available at every frame: press or wait. A hit is a press at or after the moment of the tilt; only hits are rewarded, with the amount set by the cue’s colour.

A trial is valid when the tilt occurs at the cued quadrant and invalid when it occurs elsewhere. From the agent’s choices we computed three behavioural measures. The hit rate is the fraction of trials ending in a hit, computed separately for each tilt size and for valid versus invalid trials. The reaction time is the number of frames between the tilt and the press. The validity benefit is the hit rate on valid trials minus the hit rate on invalid trials at a fixed tilt size — the behavioural signature of cue-driven attention. We also measured the hit rate separately by cue colour, to test whether the promised reward changes detection itself.

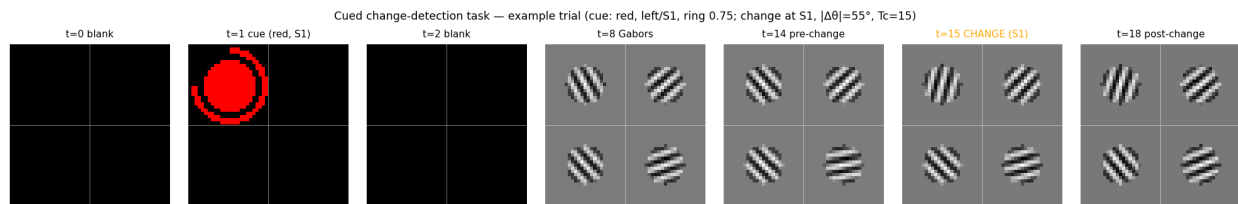


Figure 2: The cued change-detection task. A coloured ring cue (colour = reward value, completeness = advertised reliability) points to one quadrant; later, one of four gratings tilts, and the agent must press at or after the tilt.

2.2 The two networks

Both agents share one architecture: a recurrent vision network with no convolutions in its visual front end. Each 50×50 frame is cut into 100 non-overlapping patches of 5×5 pixels, arranged as a 10×10 grid, and each patch is embedded into a vector called a token. A single attention block with 8 parallel heads then lets every patch token gather information. In attention, each token acts as a query that assigns a weight to every available key; the weights are produced by a softmax (so they are positive and sum to 1), and the token receives the weighted average of the keys’ contents. Here each of the 100 patch queries sees 300 keys: the 100 patch tokens themselves (the image keys) plus two banks of 100 recurrent memory tokens (the memory keys). These memory tokens are the hidden states, H1 and H2, of two stacked long short-term memory networks (LSTMs) — recurrent units that carry information from frame to frame — with one memory slot per patch position, so memory has the same 10×10 spatial layout as the image. The attention output drives the two LSTMs, and the top memory bank H2 is read by two small one-dimensional convolutional heads: an actor that outputs the press/wait decision, and a critic that outputs its estimate of expected future reward as a distribution over 51 quantiles (51 points spanning the range of outcomes it considers possible). The networks were trained purely by reinforcement learning: the actor is improved with a maximum-a-posteriori policy optimisation loss combined with behavioural cloning (it is pulled toward actions the critic scores highly while staying close to its own recent behaviour), the critic is trained as a distributional quantile critic, past experience is replayed with priority given to surprising episodes, and a slowly updated copy of the critic (a target network) stabilises learning.

The two agents differ in exactly one connection. An attention block normally carries a skip connection (also called a residual): the block’s input is added back to its output, so information can travel around the attention rather than through it. In the baseline network, this skip connection carries the current frame’s visual input — so the image can reach the LSTMs even if no attention is paid to it. In the memory-routed network, the skip connection instead carries the previous frame’s memory H_1 — so the current image can reach the rest of the network only by being attended to. The rationale is straightforward: if vision can bypass attention, attention is free to become decorative; if vision cannot, attention is forced to do real work. Comparing the two networks therefore tests whether closing the bypass makes visual attention genuinely load-bearing. Key comparisons were repeated across three independently trained networks with different random seeds.

2.3 Reading the attention faithfully

To study attention we did not rely on approximations. We built an exact replica of the network’s attention computation that, given the same inputs, reproduces the trained network’s outputs to numerical precision (agreement at floating-point accuracy), while additionally exposing the full set of attention weights for every head, query, key, and frame, and accepting an additive bias on the attention scores before the softmax (used for the causal test in section 2.5). Because the replica is numerically identical to the network, the weights we report are exactly the weights the network used to act. From these weights we computed three descriptions. The attention budget is the fraction of each query’s weight placed on image keys versus memory keys, averaged over queries and frames. The sharpness of a query is the effective number of keys it attends to, defined as the exponential of the entropy of its weight distribution: a query that spreads its weight evenly over k keys has an effective count of k , so smaller numbers mean sharper attention. Finally, quadrant time courses track, frame by frame, how much weight falls on the keys belonging to each of the four quadrants, separately for image keys and memory keys.

2.4 Linear decoding of the latents

To ask what the network’s internal states contain, we trained simple linear decoders — linear classifiers for categorical variables and linear regressions for continuous ones — to read task variables directly from the recorded latent activity: the cue’s colour, its advertised reliability, the elapsed time within the trial, whether a change has occurred, the size of the change, and the quadrant where it occurred. All decoders used 5-fold cross-validation: trials were split into five parts, the decoder was fit on four and tested on the held-out fifth, rotating so every trial is tested once; we report held-out classification accuracy or R^2 (the fraction of variance explained). As a control, every decoder was re-fit on shuffled labels, which reduced performance to chance in all cases. To locate where information lives, we compared two ways of summarising the 100 memory tokens before decoding: the overall average (the mean of all 100 tokens, a single vector that discards spatial layout) and quadrant-wise pooling (the mean of the 25 tokens within each quadrant, four vectors concatenated, which preserves coarse layout). A variable readable from quadrant-wise pooling but not from the overall average is carried by the spatial pattern of memory, not by its mean activity.

To measure where the read-out heads themselves look, we computed gradient saliency: the gradient of the actor’s press output (and, separately, the critic’s value output) with respect to each of the 100 memory positions, taking its magnitude at each position and frame and summing within quadrants. Large saliency at a position means small changes in that memory slot strongly move

the decision.

2.5 The causal attention test

Decoding shows what information is present; it does not show what the network uses. To test use, we intervened on attention directly. The method, in one sentence: add a small constant bias to the pre-softmax attention scores of one chosen group of keys — all image keys, all memory keys, or only one quadrant’s image or memory keys — replay the task with everything else unchanged, and measure how the hit rate moves. The bias was swept from -6 to $+6$ (negative values starve the group of attention, positive values force attention onto it). Each condition used 256 paired trials: the identical trial — same cue, same change location, time, and size — was run once with and once without the bias, so any difference in outcome is caused by the bias alone. We summarise each condition by the hit-rate swing $|\Delta\text{hit}|$: the difference between the highest and lowest hit rate produced across the bias sweep. For quadrant-specific biases, the cue was held at a fixed position throughout: because the bias targets a fixed set of key positions on the screen, fixing the cue guarantees that the nudged quadrant always stands in the same known relation to the change — always the changed quadrant, or always the opposite one — rather than landing on an irrelevant location on a random subset of trials. An independent replication used a smaller bias strength (± 4) across the three training seeds.

2.6 Value and uncertainty measures

The critic’s predictions were evaluated in four ways. Calibration: for each trial we regressed the reward actually obtained on the critic’s predicted value (the mean of its 51 quantiles); a regression slope of 1 means the critic’s promises match outcomes. Value time course: the predicted value at every frame, averaged within trials grouped by cue colour and by whether a change occurred, showing when and how the network’s expectation tracks the cue’s advertised reward. Advantage of pressing: the critic values each action, and we define the advantage as the predicted value of pressing minus the predicted value of waiting at each frame; the frame at which this advantage crosses zero marks the moment the network first expects pressing to pay off, which in a well-timed agent should coincide with the change itself. Uncertainty: the spread of the critic’s 51 quantiles, summarised as the entropy of the distribution they imply, computed separately for each change size; because this is a differential entropy it can be negative when the distribution becomes very narrow, and a drop in entropy means the critic has become more certain about the trial’s outcome.

3. Results

We trained two variants of the same recurrent vision agent — the baseline network, in which the attention block’s skip connection carries the current visual input, and the memory-routed network, in which that skip connection instead carries the previous frame’s memory, so that visual information can reach the rest of the network only by being attended to — and we report here how this single change altered behaviour, attention, and the internal control of the decision.

3.1 Behaviour: fast, accurate, value-guided detection with a smaller cueing benefit

The memory-routed network learned the task well. Its detection performance follows a classic psychometric curve: the probability of a hit (a press at or after the moment a grating tilts) rises steeply with the size of the tilt, $|\Delta\theta|$. On valid trials, where the change occurs at the cued quadrant,

hit rates were 0.031 at 2°, 0.137 at 5°, 0.516 at 9°, 0.906 at 14°, 0.973 at 20°, 0.996 at 30°, 0.996 at 44°, and 1.00 at 64°. On invalid trials, where the change occurs at an uncued quadrant, the corresponding rates were 0.059, 0.070, 0.445, 0.832, 0.980, 1.00, 1.00, and 1.00. The network is also fast: for large tilts ($|\Delta\theta| \geq 44^\circ$) the median reaction time is 0 frames — the press lands on the very frame of the change — and 93% of all hits occur within 1 frame of the change. The baseline network needed 1–2 frames for the same decisions.

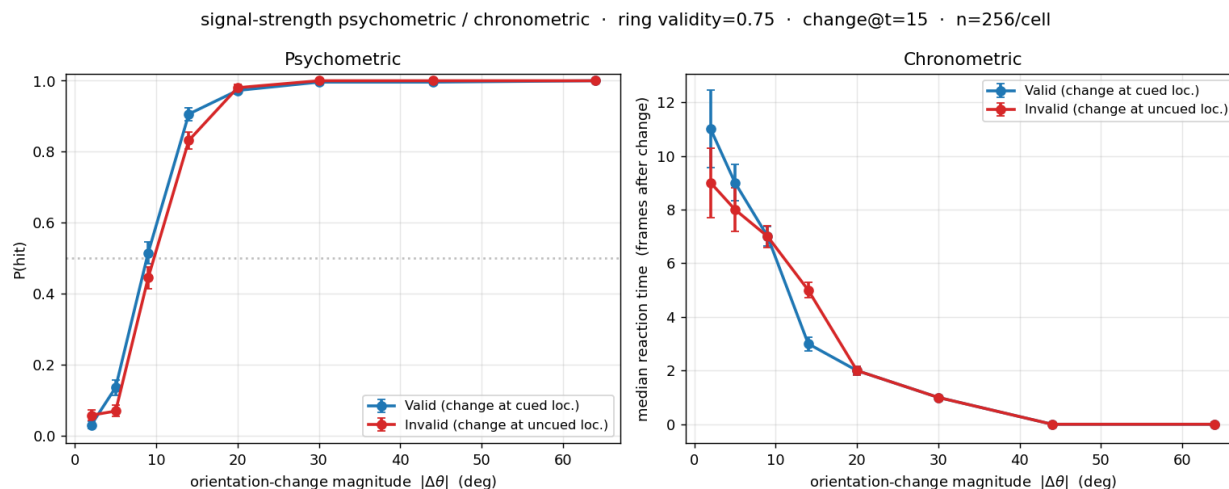


Figure 3: Detection accuracy and speed. Hit rate rises steeply with tilt size on both valid (cued) and invalid (uncued) trials, and reaction times shrink to a median of 0 frames for large tilts.

Cueing still helps, but less than in the baseline network. We measured the validity benefit — the hit rate on cued-location changes minus the hit rate on uncued-location changes — near the detection threshold, where the cue matters most. The memory-routed network shows a benefit of +0.084 at 8° and +0.112 at 9° (consistent across independently trained networks). This is roughly half the baseline network’s benefit of $\approx +0.18$. Forcing all visual information through attention therefore came at a measurable behavioural cost: the network leans less heavily on the spatial cue.

A second feature of the cue, however, is ignored entirely. The completeness of the cue ring advertises how reliable the cue is, from 25% to 100%. Performance is flat across this entire range (hit rates between 0.34 and 0.45 near threshold, with small fluctuations that do not replicate across networks). The network reads the cue’s location but does not modulate its behaviour by the cue’s advertised trustworthiness.

In contrast, the cue’s value — its colour, which signals the reward for a correct response (red 5, green 3, blue 1) — strongly shapes behaviour. At a near-threshold tilt of 8°, the network detects the change on only 0.25 of blue-cued trials, versus 0.48 for green and 0.44 for red. The lowest-value cue is detected far worse, while the two higher values are treated almost identically. Detection effort, in other words, tracks what a hit is worth.

3.2 An image-watching attention head emerges

The single attention block gives every patch token a choice among 300 keys: the 100 image keys (the current frame’s patch tokens) and the 200 memory keys (the two banks of recurrent memory tokens). Averaged over a trial, the memory-routed network places 17.6% of its attention on image keys and 82.4% on memory keys — memory still dominates the budget. But that average hides

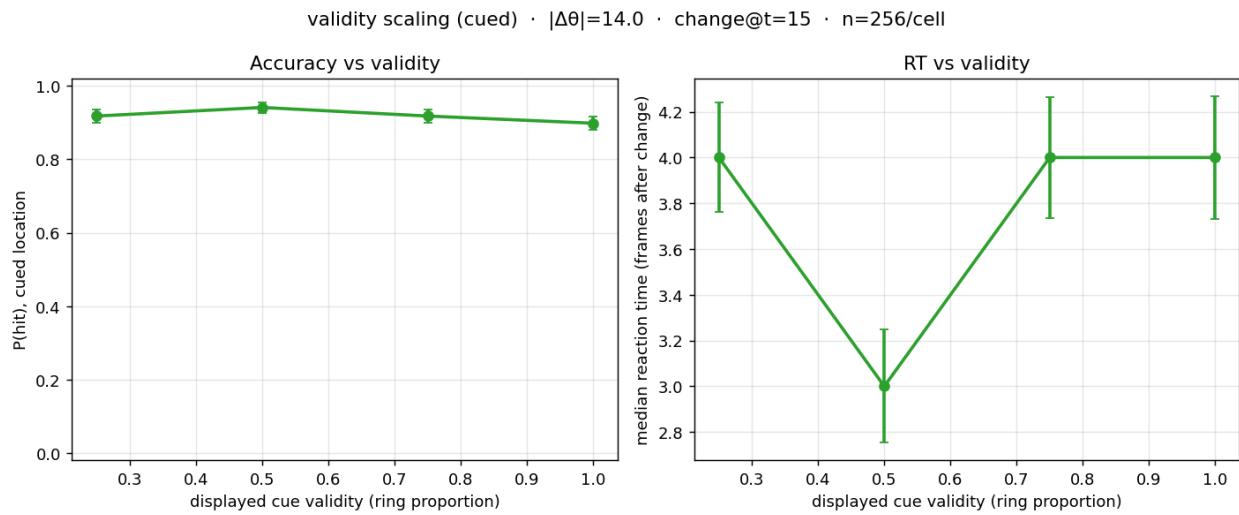


Figure 4: Advertised cue reliability does not change performance. Hit rate near threshold is flat as the cue's advertised reliability sweeps from 25% to 100%.

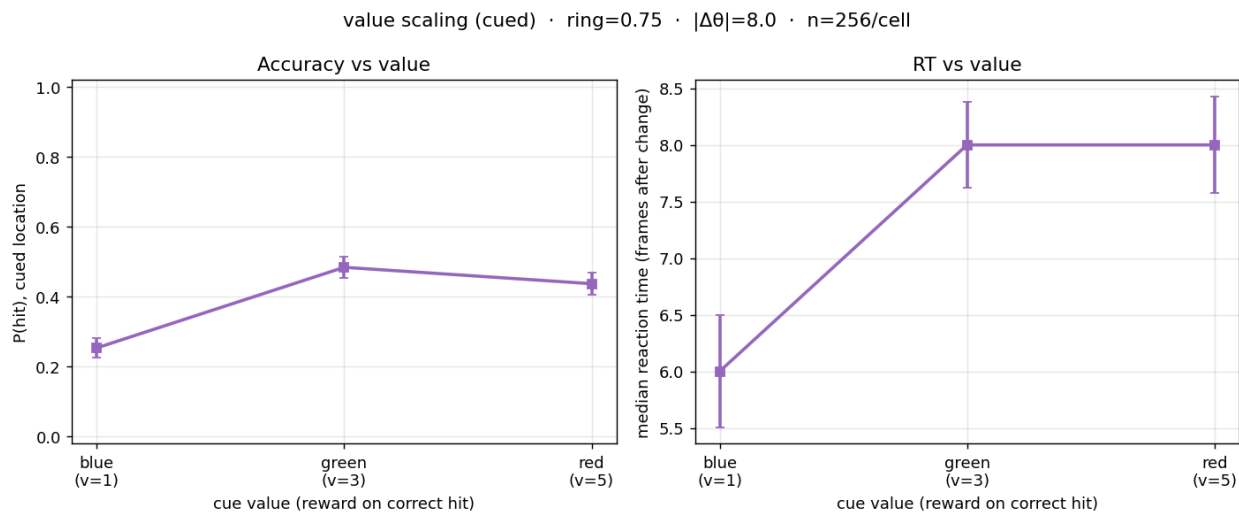


Figure 5: Detection follows the cue's value. At a near-threshold tilt, low-reward blue cues yield far fewer hits (0.25) than green (0.48) or red (0.44) cues.

a sharp division of labour: one head (head 0) devotes 58.8% of its attention to image keys, more than twice the next-highest head (25.3%). It is, in effect, a dedicated image-watcher. No such head exists in the baseline network, where the mean attention on image keys is only 9.5% and even the most image-leaning head reaches only about 20%. When the skip connection no longer delivers the image for free, the network builds a head whose job is to bring it in through attention.



Figure 6: A dedicated image-watching head. Per-head maps of attention on the image keys; head 0 concentrates the majority of its attention on the current frame, unlike any head in the baseline network.

Attention is also far more selective. For 6 of the 8 heads, a typical query concentrates its weight on the equivalent of only about 15–21 of the 300 available keys (the remaining two heads are broad, spreading over about 60). In the baseline network the same measure ranges from about 38 to 103 keys — attention there is closer to a blur than a spotlight.

This sharpened attention behaves like task-directed attention should. The attention on image keys orients to the cued quadrant the moment the cue appears and follows the cue’s side throughout the trial — a top-down, cue-driven signal.

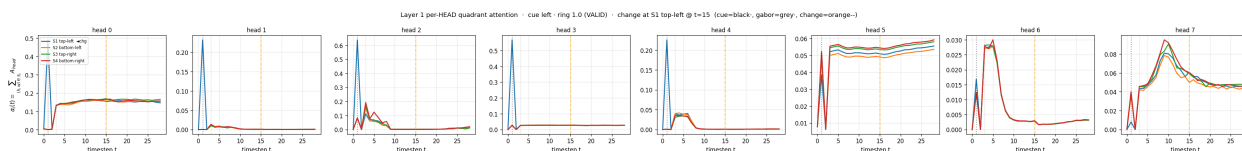


Figure 7: Image attention follows the cue. Attention on image keys, split by quadrant over the course of a trial: it shifts to the cued quadrant at cue onset and tracks the cue’s side.

The attention on memory keys plays the complementary role: it locks onto the changed quadrant

at the moment the change occurs — a bottom-up, event-driven signal.

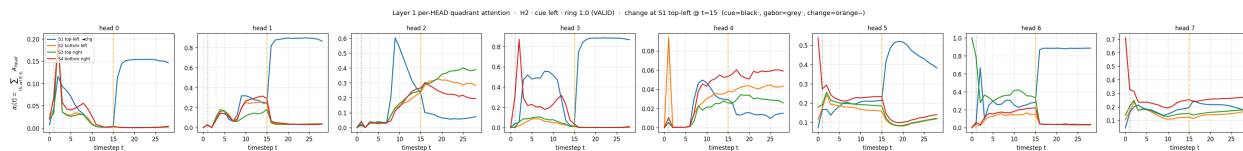


Figure 8: Memory attention locks onto the change. Attention on the top memory bank, split by quadrant: it concentrates on the changed quadrant from the moment of the change.

Together, the two key types implement a clean division: image attention carries the cue’s prediction of where to look, while memory attention reports where something actually happened.

3.3 Where the network keeps what it knows

To find out which facts about a trial the network actually stores, we trained simple linear readouts — each one a single weighted sum of the network’s internal numbers, fitted to predict one feature of the trial. A readout that succeeds tells us that the feature is present in the network’s memory in a directly usable form. Here and throughout this section we study the memory-routed network unless we say otherwise. We score each readout either by accuracy (the fraction it gets right, where 1.00 is perfect) or, for continuous quantities, by R^2 (the fraction of the variation it explains, where 1.00 is again perfect).

Two kinds of information are held almost flawlessly. The cue’s colour and its advertised reliability — the completeness of the ring that signals how trustworthy the cue is — are both read out perfectly (1.00). Elapsed time within the trial is read out almost perfectly ($R^2=0.998$), so the network keeps an accurate internal clock. Facts about the change itself are held more weakly but clearly: whether a change has happened yet reads out at 0.78, and how large the tilt was reads out at 0.67.

The most informative result concerns where the change occurred. The four gratings sit in four quadrants, and the network has one memory slot per patch position, so memory has a spatial layout that mirrors the image. When we pool the top memory bank quadrant by quadrant — keeping that layout — the location of the change reads out at 0.93. When we instead collapse the same memory into a single overall average, location reads out at only 0.43. The information is therefore carried by the spatial arrangement of memory, not by its average level: the network knows where the change is because the relevant memory slots light up in the right place, and averaging across positions throws that away.

The features computed by the two read-out heads — the actor that decides press or wait and the critic that estimates reward — also carry the change location strongly, at 0.90–0.91. The baseline network, whose skip connection feeds the current image around the attention block rather than the previous memory, carries the same information in its read-out features less well, at 0.81–0.83.

3.4 Re-routing one connection redistributes causal control

Decoding shows what information is present, but not what the network actually uses to decide. For that we ran a causal test. We added a small bias to the attention scores of one group of keys — either the 100 image keys, all 200 memory keys, or just the keys belonging to a single quadrant — then replayed the task and measured how much the hit rate moved. Each condition was a paired comparison on the same 256 trials, with the bias swept from -6 to $+6$, and the cue held at a fixed

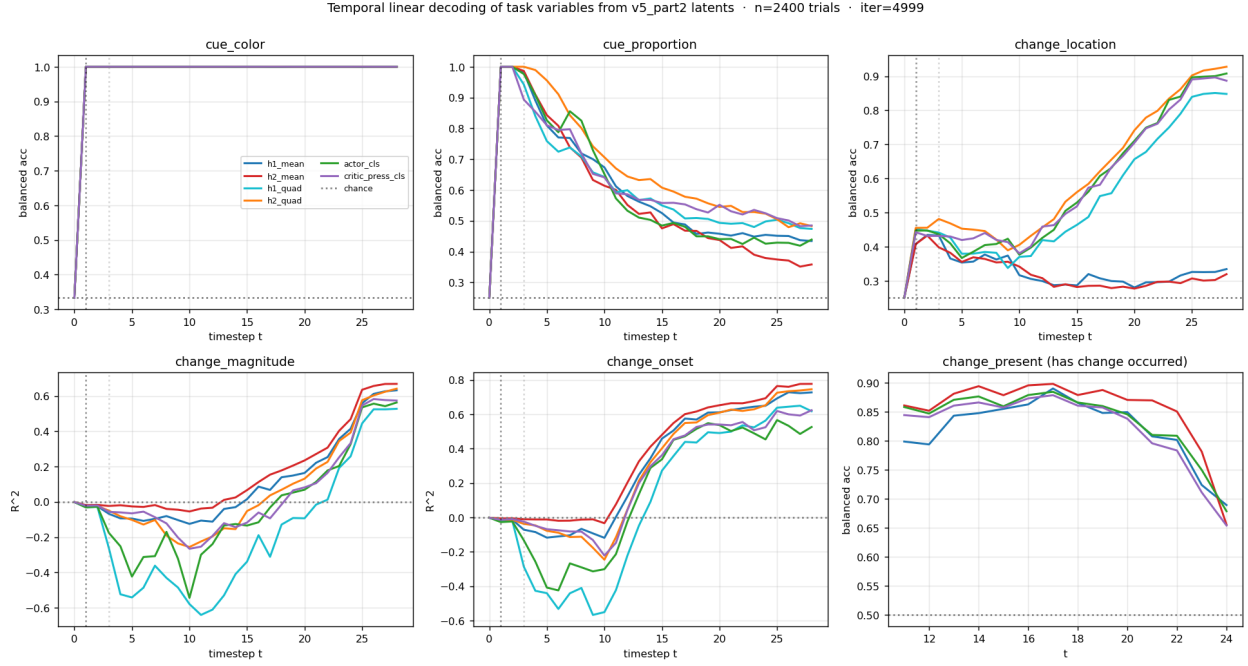


Figure 9: What the network stores and where the change location lives. Left: how well simple linear readouts recover each task feature over the course of a trial. Right: the change location is recovered far better from memory kept in its spatial layout (0.93) than from its overall average (0.43).

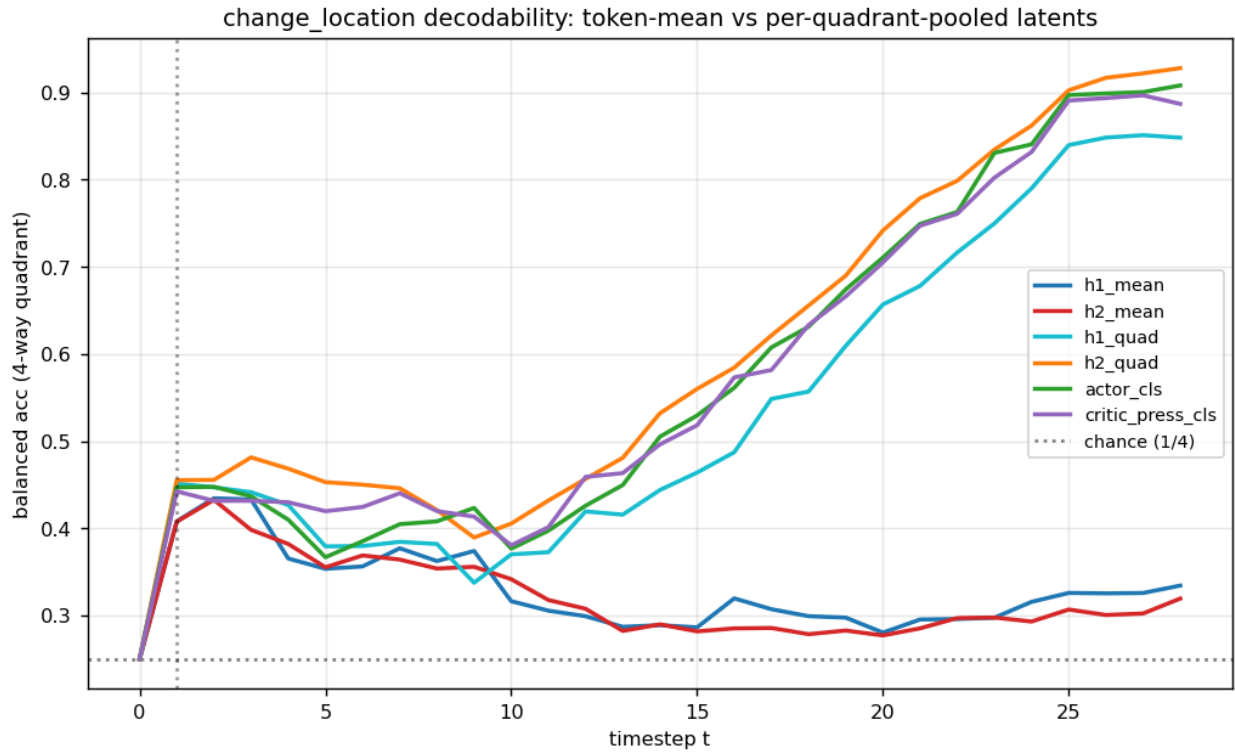


Figure 10: Change location is held in the layout of memory, not its average. Reading the change location from quadrant-by-quadrant pooling of the top memory succeeds (0.93); reading it from a single average across positions largely fails (0.43); the read-out heads' own features carry it at 0.90–0.91.

position so that quadrant-specific nudges always pointed at the spot that mattered. The size of the resulting hit-rate swing, written $|\Delta\text{hit}|$, tells us how much control over the decision flows through that group of keys.

Start with the baseline network. Here control is concentrated in one place. Biasing the memory keys of the changed quadrant alone swings the hit rate by 0.363 on valid trials — trials where the change happens at the cued quadrant — and by 0.496 on invalid trials, where the change happens away from the cue. Nudging the image keys, by contrast, moves the hit rate by only 0.070, and nudging all memory keys together by 0.090. In other words, the baseline network’s decision is effectively hostage to a single memory hotspot: poke that one spot and behaviour lurches; poke the image and almost nothing happens.

The memory-routed network behaves completely differently. No single knob dominates. On valid trials the largest swing now comes from the image keys, at 0.090 — slightly more than all memory keys together at 0.086. The changed-quadrant image keys move the hit rate by 0.059 and the changed-quadrant memory keys by 0.051, with the opposite quadrant’s image keys weaker still at 0.027. The memory hotspot that ruled the baseline network has been cut down from 0.363 to 0.051, a roughly seven-fold reduction, while the image pathway has risen to share control.

The invalid trials make the contrast sharpest. When the change appears at the top-left quadrant while the cue points at the bottom-right, the changed-quadrant memory keys swing the hit rate by 0.148 — the strongest single effect under conflict, as expected, since memory is where the off-cue change must be registered. But this is far below the baseline network’s 0.496 for the same manipulation. Biasing all memory keys together moves the hit rate by 0.082, and inspection shows this mostly shifts the critic’s value estimate rather than the press choice itself; the image keys move behaviour by only 0.027. Across both trial types, re-routing the skip connection cut the memory hotspot’s grip by between three- and seven-fold (0.363→0.051 on valid trials, 0.496→0.148 on invalid trials) and lifted the image pathway to co-dominant status (0.090, at or above all-memory’s 0.086). A separate run at a gentler bias strength (± 4 , three seeds) reproduced the same ordering at matching scale, with the memory hotspot at 0.066 and the image pathway at 0.038.

The two networks reach almost the same behaviour, yet they reach it through opposite internal arrangements: in the baseline network one memory hotspot holds a near-monopoly over the decision, while in the memory-routed network control is shared across a balance of image and memory pathways. The same outward performance can be produced by an internal monopoly or by a balanced coalition, and the single choice of where the skip connection runs decides which one the network builds.

3.5 The read-out follows the change

Finally we asked what the actor — the head that issues the press — is actually watching when it decides. We used gradient saliency, which measures how strongly each memory position influences the press output: positions with high saliency are the ones the actor leans on. In the memory-routed network the actor locks onto the changed quadrant once the change has happened, giving it roughly twice the saliency of the other three quadrants. This holds even on invalid trials, where the change is away from the cue: when the change is at the top-left quadrant the actor’s saliency there reaches about 24 against 12–15 elsewhere. The actor is following the change, not the cue.

This is a further point of divergence from the baseline network. There, the actor’s saliency was diffuse — spread across positions rather than fixed on the changed quadrant — and only the critic, the head that estimates reward, was reliably change-locked. Re-routing the skip connection there-

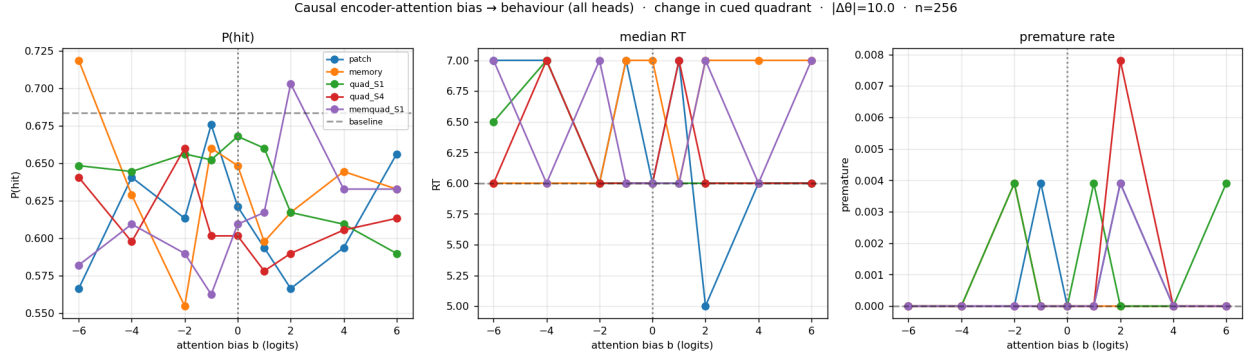


Figure 11: One wiring choice changes who controls the decision. Hit-rate swings from biasing different groups of attention keys. In the baseline network a single changed-quadrant memory hotspot dominates (0.363 valid, 0.496 invalid); in the memory-routed network the swings are small and shared between image and memory pathways.



Figure 12: Image and memory become co-dominant after re-routing. Side-by-side hit-rate swings for image keys versus memory keys. The memory-routed network's image pathway (0.090) matches or exceeds its full memory pathway (0.086), unlike the baseline network's memory-dominated profile.

fore did more than redistribute causal control; it pulled the press decision itself onto the location of the change.

Layer 1 per-HEAD quadrant attention · saliency (h0=actor,h1=critic) · cue left · ring 1.0 (VALID) · change at S1 top-left @ t=15 (cue=black, gabor=grey, change=orange--)

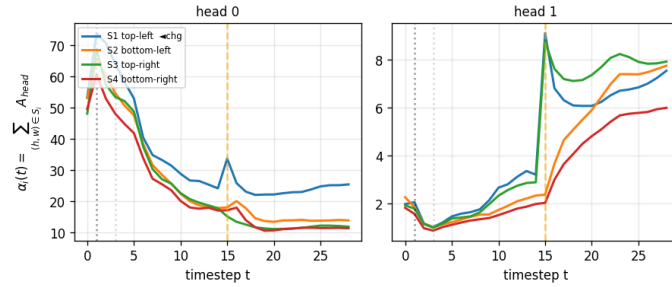


Figure 13: The actor watches the changed quadrant on valid trials. Saliency of the actor and critic heads across the four quadrants. After the change, the actor’s influence concentrates on the changed quadrant at about twice the level of the others.

Layer 1 per-HEAD quadrant attention · saliency (h0=actor,h1=critic) · cue right · ring 1.0 (INVALID) · change at S1 top-left @ t=15 (cue=black, gabor=grey, change=orange--)

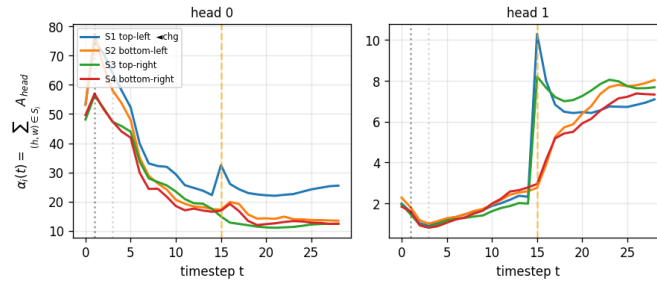


Figure 14: The actor follows the change even against the cue. On invalid trials, where the change is away from the cued quadrant, the actor’s saliency still locks onto the changed quadrant (about 24 versus 12–15 elsewhere), unlike the baseline network’s diffuse actor.

3.6 A calibrated, value-guided critic that commits at the moment of evidence

The critic is the part of the memory-routed network that predicts how much reward is coming. We asked whether those predictions are accurate, whether they track the value advertised by the cue, and whether they tell the actor when to press. On all three counts the answer is yes.

First, accuracy. We compared the critic’s predicted value against the reward each trial actually delivered. The two line up almost perfectly: a regression of obtained reward on predicted value has a slope of 1.053 and explains most of the variance ($R^2 = 0.660$). The critic is not merely ranking trials — it reports reward on the correct numerical scale.

Second, the predictions are ordered by the cue’s value from the instant the cue appears. The coloured ring tells the agent what a correct response is worth (red 5, green 3, blue 1). At cue onset the critic’s predicted value already separates by colour: 3.76 for red, 2.32 for green, and 0.86 for blue. As the trial unfolds, each prediction climbs toward the full reward at the moment the grating tilts. Trials in which no grating ever changes stay flat at roughly 2.3, the value of an outcome that is still undecided. The critic therefore carries the cue’s economic meaning forward across the blank frames and updates it only when sensory evidence arrives (Fig. exp5_value_timecourse).

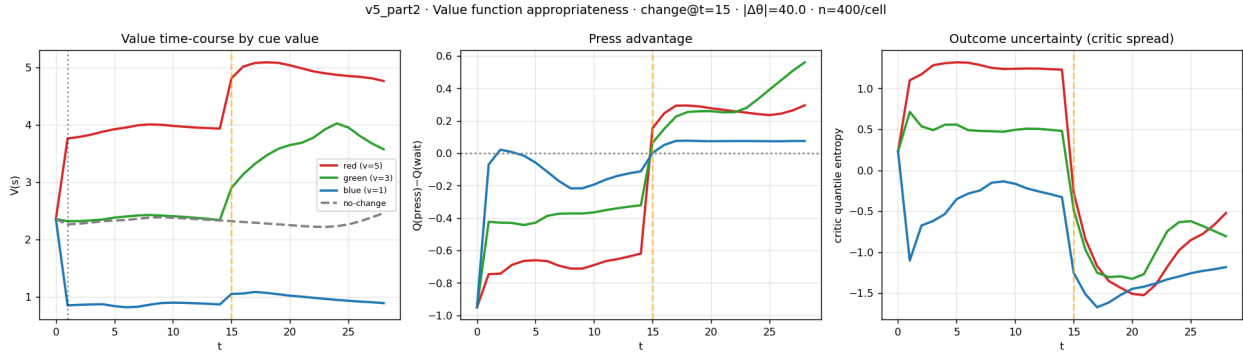


Figure 15: The critic’s value forecast tracks the cue’s worth and rises at the change. Predicted value over the trial, separated by cue colour; red, green and blue cues start at different levels set by their reward and climb toward the full payout when the grating tilts, while no-change trials stay flat.

Third, and most useful for behaviour, the critic tells the actor when pressing becomes worthwhile. We computed the predicted advantage of pressing over waiting and asked when it crosses zero — the point at which the network expects acting to beat holding. For red and green cues the crossing falls exactly at the change frame: the network judges a press to be worth it the moment the evidence is present, not before and not after. This is a clear improvement over the baseline network, whose press-advantage crossed zero a full 5 to 8 frames after the change — a built-in hesitation. Re-routing vision through attention removed that lag and let the agent commit at the moment of evidence, which is why its reaction times are so short (a median of 0 frames for easy changes). The one exception is the blue cue: its press-advantage hovers near zero throughout the trial and never crosses cleanly. We flag this as a genuine caveat rather than a feature — it is consistent with the agent’s weak behaviour on blue (low-value) cues, where a correct response is barely worth more than doing nothing, so the value signal that should drive the press is too small to separate from noise.

Finally, the critic’s confidence grows as the evidence does. The critic does not output a single number; it outputs a distribution over 51 possible reward levels (a set of quantiles), and the spread of that distribution is a read-out of how uncertain the network is. As the change becomes easier to see, this spread collapses by roughly threefold: an uncertainty measure of 0.45 when no change occurs falls to -1.02 for the largest, most obvious tilts (Fig. exp5_entropy_by_mag). The same sharpening is visible when we relate the critic’s confidence to where attention is pointing (Fig. exp5_calibration_attn): the network is most certain exactly when its attention has locked onto the changed location. The value system is thus calibrated in both senses — its mean prediction matches real reward, and its spread shrinks as the world becomes more predictable.

4. Discussion

4.1 What one skip connection buys

The two networks we studied are identical in every respect but one. In the baseline network, the attention block has a shortcut, or skip connection, that carries the current camera image around the attention computation, so visual information can reach the rest of the agent whether or not attention selects it. In the memory-routed network, that same shortcut instead carries the previous frame’s memory, which means fresh visual information has only one way in: it must be attended to. Changing what that single wire carries reshaped the agent in five connected ways.

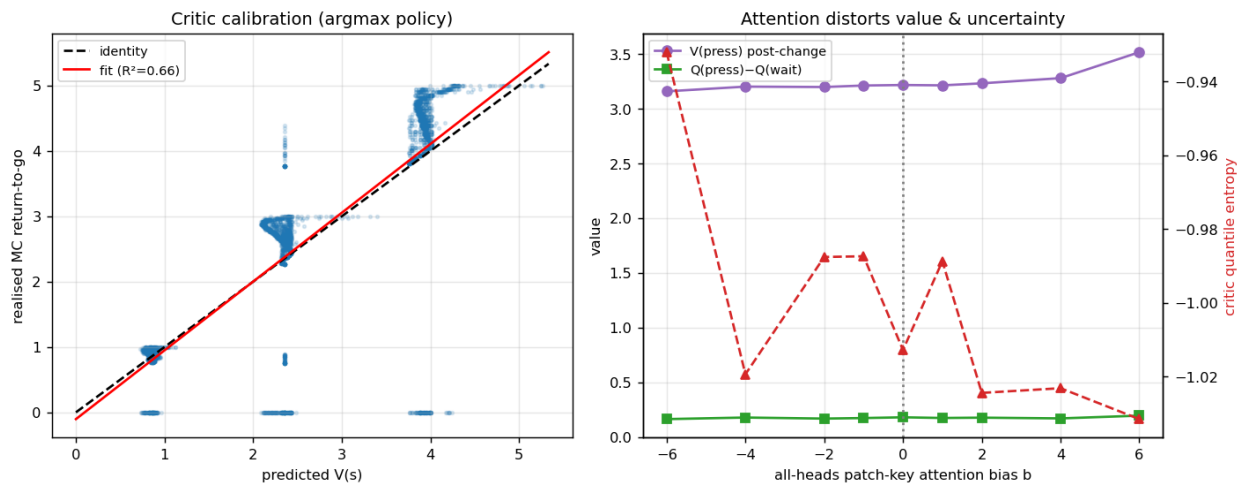


Figure 16: Confidence and attention move together. The critic becomes more certain (a narrower predicted reward distribution) precisely on trials where attention has concentrated on the relevant location.

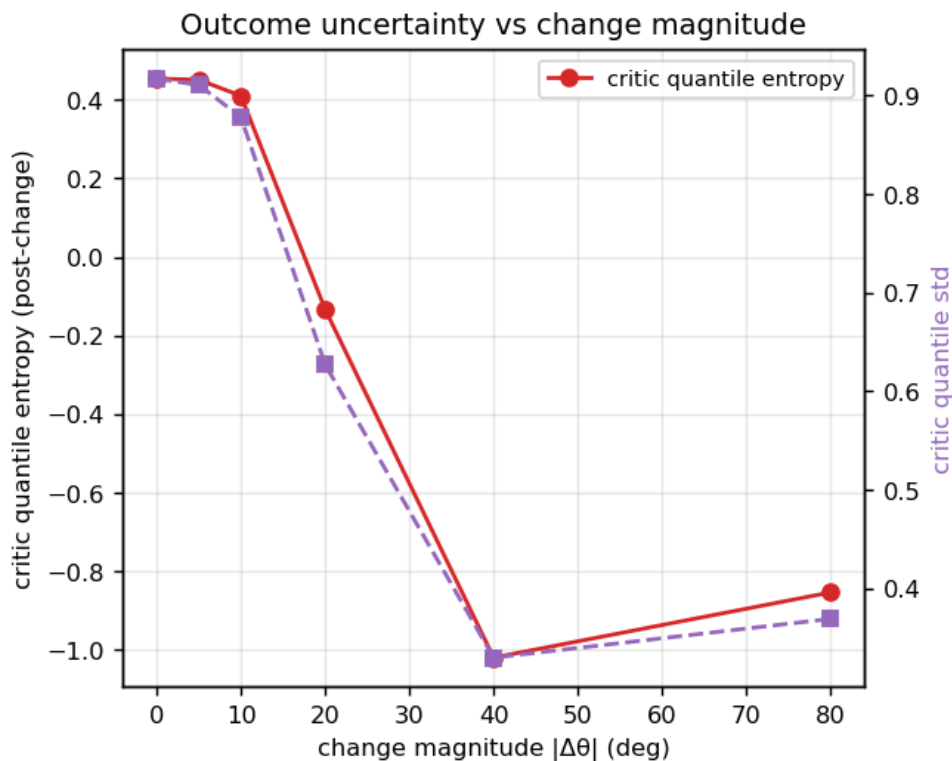


Figure 17: Uncertainty collapses as the change gets easier to see. The spread of the critic's 51-quantile forecast falls about threefold from no-change trials to the largest tilts.

It created a dedicated sensory channel. In the memory-routed network one of the eight attention heads became a specialist image-watcher, placing 58.8% of its attention on the image (the next-busiest head reaches only 25.3%). In the baseline network no head ever specialised on the image at all (the average head spent 9.5% of its attention there, the most image-focused around 20%). When vision is the only thing that needs attending, the network spends an attention head on it.

It sharpened attention. A typical query in the memory-routed network concentrates on roughly 15 to 21 of its 300 available keys for six of the eight heads, against 38 to 103 keys in the baseline. Forcing information through the attention bottleneck made that bottleneck precise.

It sped up commitment. Because the critic's press-advantage now crosses zero at the change rather than 5 to 8 frames later, the agent acts almost immediately on easy trials (median reaction time of 0 frames; 93% of hits within one frame of the change), where the baseline needed one to two frames.

It spatialised the read-out. The actor now locks onto the changed quadrant after the change — about twice the influence of any other quadrant, even on invalid trials where the change is away from the cued location. In the baseline the actor's read-out was diffuse and only the critic was change-locked; here the decision-making head itself knows where the change was.

And, most importantly, it broke a monopoly. In the baseline network the entire decision was hostage to a single memory hotspot: nudging the attention paid to the changed quadrant's memory keys swung the hit rate by 0.363 on valid trials and by 0.496 on invalid trials. After re-routing, that same nudge moves the hit rate by only 0.051 and 0.148 — a three- to seven-fold loss of grip. In its place the image pathway rose to co-dominance: nudging the image keys now moves behaviour by 0.090, equal to or larger than the 0.086 from all memory keys combined. Control over the decision was redistributed from one internal hotspot to a genuine balance of incoming vision and stored memory. This is exactly the outcome the architecture was designed to force, and it confirms that the change is causal, not cosmetic.

4.2 The price: a weaker cue prior

Honesty requires reporting what was lost. The cue in this task is informative: a valid cue points at the location most likely to change, and a good agent should use it as a prior, detecting cued changes more readily than uncued ones. Both networks show this validity benefit, but the memory-routed network shows about half as much of it. Near the detection threshold the cued-minus-uncued advantage is +0.084 at an 8° tilt and +0.112 at 9°, against roughly +0.18 for the baseline.

We read this as a direct trade-off rather than a defect. The baseline network's large validity benefit and its single dominant memory hotspot are two views of the same fact: that network leaned heavily on an internal prior stored in memory, which let the cue strongly bias the decision but also left the decision captured by one internal variable. The memory-routed network, by forcing vision to compete on equal footing, necessarily lets the stored cue prior count for less. A more balanced, sensory-grounded decision and a strong top-down prior pull in opposite directions; pushing vision into the loop bought the former at the cost of the latter. Notably, what the cue still controls strongly is value, not spatial priority: near threshold a low-value blue cue is detected far worse (0.25) than a green (0.48) or red (0.44) cue, while the cue's advertised reliability has no measurable effect on performance (a flat 0.34–0.45 across the full 25–100% range). The agent listens to what a correct response is worth far more than to how reliable the cue claims to be.

4.3 Implications for attention research

Two general lessons follow. The first concerns the difference between attention that is imposed and attention that is earned. Both networks have exactly the same attention machinery, but only when we removed the visual shortcut did attention become genuinely load-bearing — specialised, sharp, and causally in control of behaviour. A model can contain a perfectly good attention mechanism and still route most of its real computation around it. Demonstrating that a system has attention is therefore not the same as demonstrating that attention does the work; architecture determines which.

The second lesson is methodological. We could only see the redistribution of control by intervening — adding a small bias to one group of attention keys, replaying the trials, and measuring how behaviour moved. Correlational read-outs alone would have missed it: the baseline and memory-routed networks both look attentive in their maps. Crucially, the intervention had to be aligned with the geometry of the trial. We held the cue at a fixed position so that a nudge aimed at a particular quadrant always landed on the spot that actually mattered. A quadrant-specific intervention that ignored where the relevant change was would have averaged its own effect away. Causal tests of attention need to be pointed at the place the task makes important, not applied blindly.

4.4 Limitations and future directions

Several limitations bound these claims. The blue (low-value) cue is the clearest: behaviour on it is poor and its press-advantage never crosses zero cleanly, so our value-timing results rest mainly on the red and green cues. The agent also ignores advertised cue reliability entirely, which is either a strategy (value is a better guide than a self-reported reliability number) or a failure to learn that dimension; our data cannot yet separate the two. The task itself is deliberately small — 50×50 frames, four locations, a single change per trial — so the precise numbers should be read as existence proofs about architecture, not as performance benchmarks. And although the causal interventions are tightly controlled, they perturb attention scores rather than the underlying representations, so they speak to how much each pathway gates the decision, not to the full content of what each pathway computes. Natural next steps include richer scenes with multiple simultaneous changes, manipulations that make reliability pay so we can test whether the agent can learn to use it, and interventions that edit the memory contents directly to complement the attention-gating tests reported here.

5. Conclusion

We trained two versions of a convolution-free recurrent vision agent that differ in a single wire: whether the shortcut around its one attention block carries the current image or the previous frame’s memory. Forcing vision to pass through attention, rather than letting it slip around, transformed the agent. It grew a dedicated sensory attention head, sharpened its attention several-fold, learned to commit at the exact moment evidence appeared, located the change in its decision-making read-out, and — most importantly — broke the monopoly that a single internal memory hotspot had held over its choices, redistributing control to a balance of vision and memory. The cost was a halved reliance on the cue as a spatial prior, a trade-off we report plainly. The broader point is that having an attention mechanism and using one are different things: a one-line change to where information is allowed to flow decided whether the agent’s attention was decorative or load-bearing, and only a geometry-aware causal test could tell them apart.